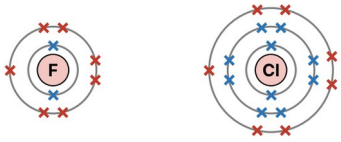
Group 7 – the Halogens

AQA GCSE Combined Science: Trilogy · Chemistry Topic 1 · Revision activity 7 · 20 marks · about 25 minutes

Read the key information (5 min), answer the **quick recall** questions without looking back (5 min), then answer the **exam questions** (15 min). Mark your work with the mark scheme on the last page.

Key information

- The group 7 elements are called the **halogens**: fluorine, chlorine, bromine and iodine. They are **non-metals**.
- They all have **seven electrons in their outer shell**, so they react in similar ways.
- They exist as **molecules made of pairs of atoms** (diatomic): F_2, Cl_2, Br_2, I_2.
- Going down the group the **relative molecular mass increases**, so the **melting and boiling points increase**.
- They react with metals to form **ionic** compounds (salts), e.g. sodium + chlorine → sodium chloride. They react with non-metals to form **covalent** compounds, e.g. hydrogen chloride.
- **Reactivity decreases going down the group.**



fluorine 2,7 chlorine 2,8,7

The halogens at room temperature

Halogen	Formula	State	Colour
fluorine	F_2	gas	pale yellow
chlorine	Cl_2	gas	green
bromine	Br_2	liquid	red-brown (orange in solution)
iodine	I_2	solid	grey-black (brown in solution)

Why does reactivity decrease down group 7?

Halogens react by **gaining one electron**. Going down the group the atoms have **more shells**, so the outer shell is **further from the nucleus**. The attraction is **weaker**, so it is **harder to gain an electron**.

Displacement reactions

A **more reactive** halogen will **displace** a **less reactive** halogen from a solution of its salt.

chlorine + potassium bromide → potassium chloride + bromine $Cl_2 + 2KBr \rightarrow 2KCl + Br_2$

The colourless solution turns **orange** because bromine has been made.

Results of mixing halogen solutions with potassium halide solutions

Halogen added	potassium chloride	potassium bromide	potassium iodide
chlorine water	—	turns orange (bromine made)	turns brown (iodine made)
bromine water	no reaction	—	turns brown (iodine made)
iodine solution	no reaction	no reaction	—

Section A: Quick recall (7 marks – 1 mark each)

1 What name is given to the group 7 elements?

2 How many electrons are in the outer shell of a halogen atom?

3 What does 'diatomic' mean?

4 What is the state of bromine at room temperature?

5 Which is the most reactive halogen?

6 What happens to the boiling point going down group 7?

7 Name the salt formed when potassium reacts with iodine.

Group 7 – the Halogens *(continued)*

Section B: Exam questions

01 Chlorine water is added to colourless potassium bromide solution.

01.1 Describe what you would see.

[1 mark]

01.2 Explain why a reaction happens.

[2 marks]

01.3 Write a word equation for the reaction.

[1 mark]

02 Bromine water is added to potassium chloride solution. Explain why there is no reaction.

[2 marks]

03 Write a balanced symbol equation for the reaction between chlorine and potassium iodide. The products are potassium chloride and iodine.

[2 marks]

04 Explain why chlorine is more reactive than bromine.

[3 marks]

05 Astatine is below iodine in group 7. Predict the state of astatine at room temperature and how its reactivity compares with iodine.

[2 marks]

Mark Scheme – Group 7 – the Halogens

(1) = one mark. / separates acceptable alternatives. Words in brackets are not needed for the mark.

Question	Marking guidance	Comments	Mark
A	**1** the halogens **2** 7 **3** a molecule made of two atoms **4** liquid **5** fluorine **6** it increases **7** potassium iodide	1 mark each	7
01.1	solution turns orange / yellow-orange (1)		1
01.2	chlorine is more reactive than bromine (1) so chlorine displaces bromine from potassium bromide (1)		2
01.3	chlorine + potassium bromide → potassium chloride + bromine (1)		1
02	bromine is less reactive than chlorine (1) so bromine cannot displace chlorine (1)		2
03	$Cl_2 + KI \rightarrow KCl + I_2$ – correct formulae (1) $Cl_2 + 2KI \rightarrow 2KCl + I_2$ – balanced (1)		2
04	chlorine has fewer shells / its outer shell is closer to the nucleus (1) so there is stronger attraction between the nucleus and an incoming electron (1) so chlorine gains an electron more easily (1)		3
05	solid (1) less reactive than iodine (1)		2

Total: 20 marks